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7MA2

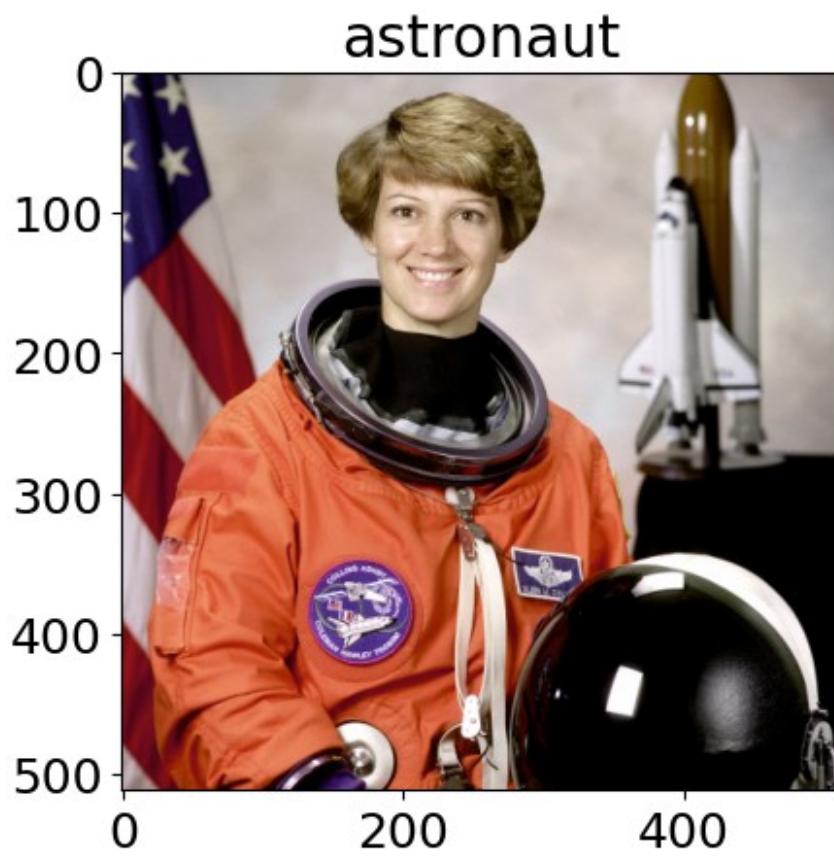
```
import matplotlib.pyplot as plt
import matplotlib
import numpy as np
import skimage as ski

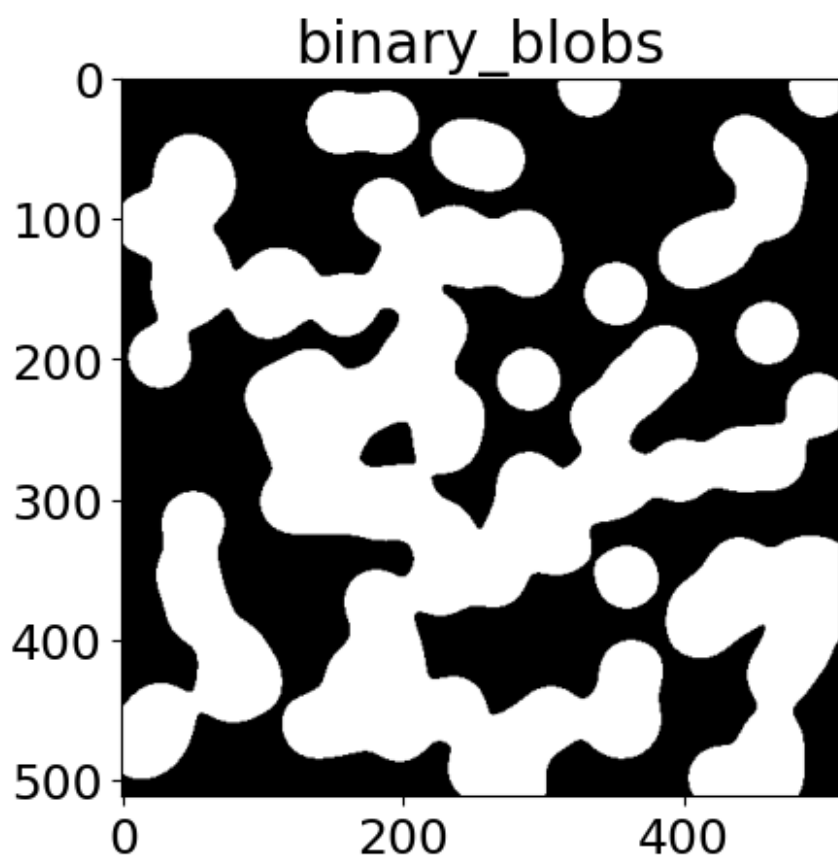
matplotlib.rcParams['font.size'] = 18

images = (
    'astronaut',
    'binary_blobs',
    'brick',
    'colorwheel',
    'camera',
    'cat',
    'checkerboard',
    'clock',
    'coffee',
    'coins',
    'eagle',
    'grass',
    'gravel',
    'horse',
    'logo',
    'page',
    'text',
    'rocket',
)

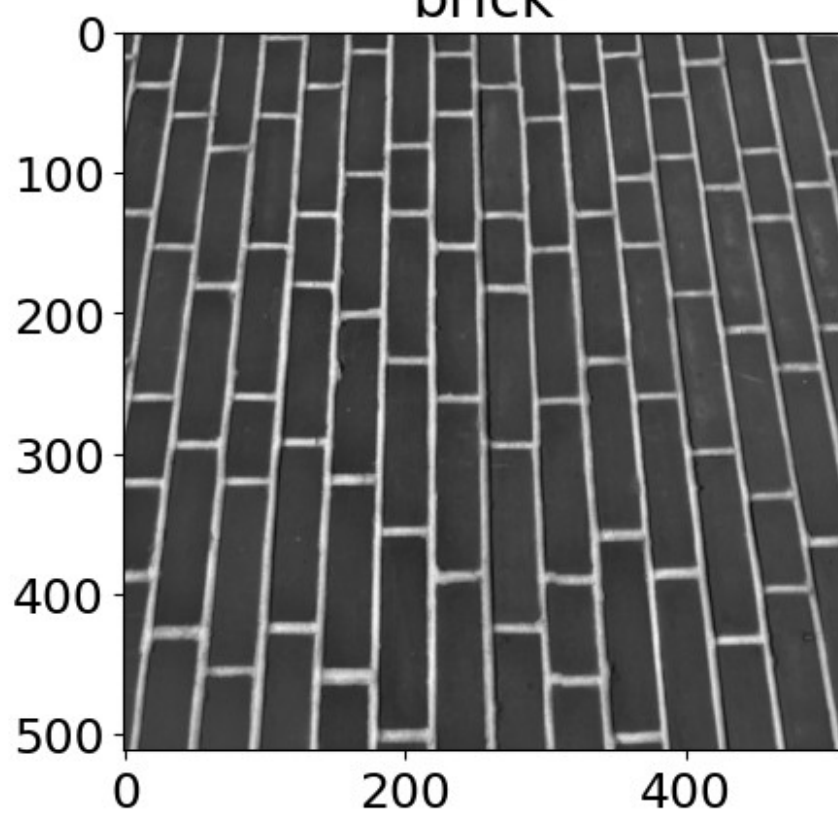
for name in images:
    caller = getattr(ski.data, name)
    image = caller()
    plt.figure()
    plt.title(name)
    if image.ndim == 2:
        plt.imshow(image, cmap=plt.cm.gray)
    else:
        plt.imshow(image)
```

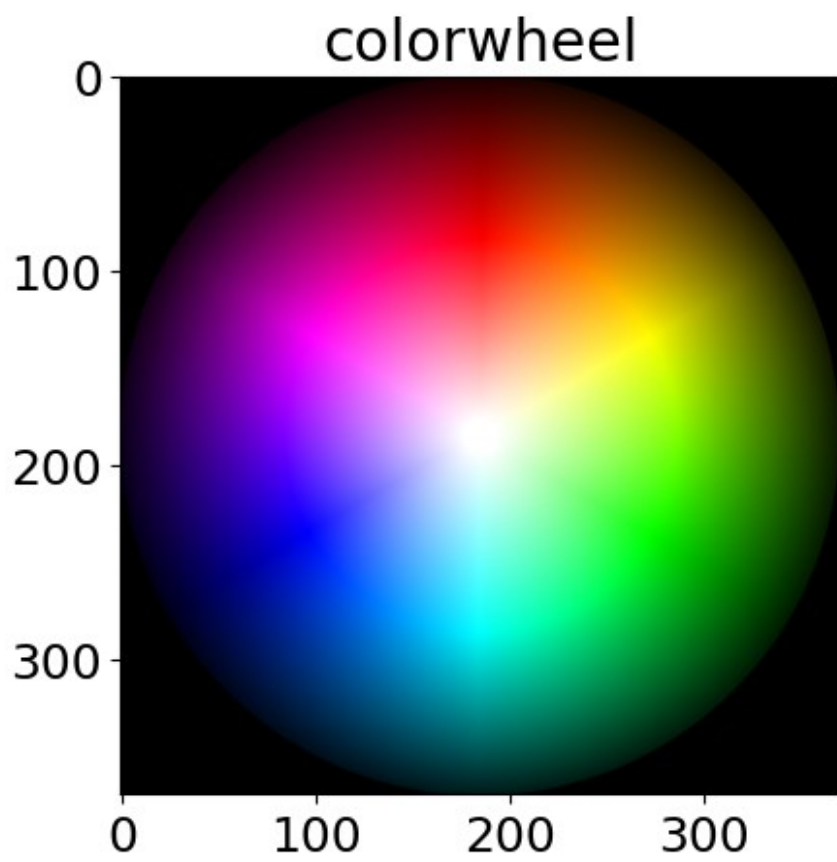
```
plt.show()
```



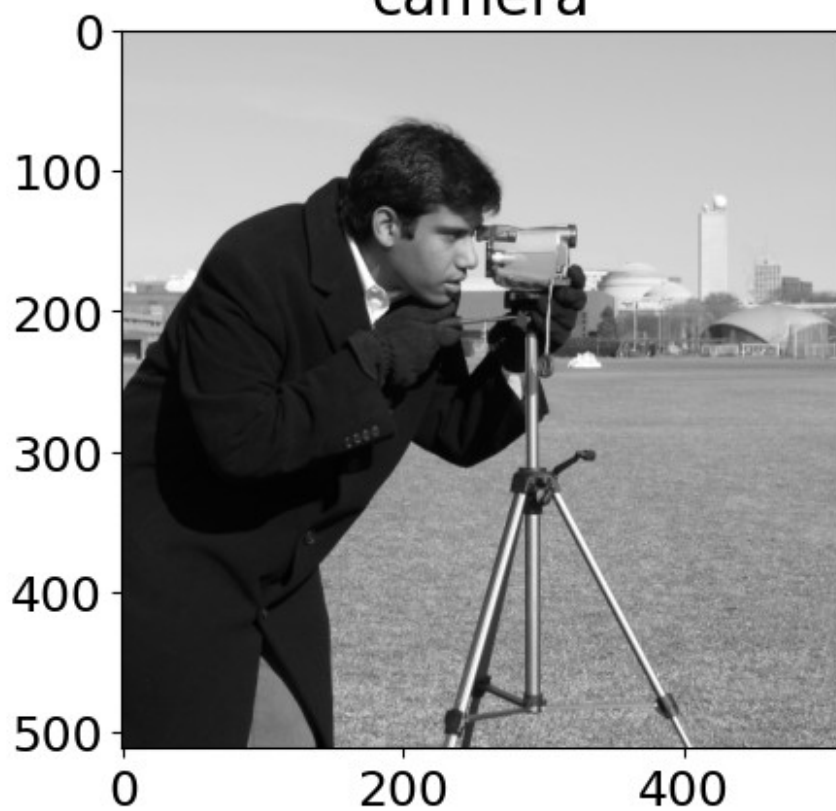


brick

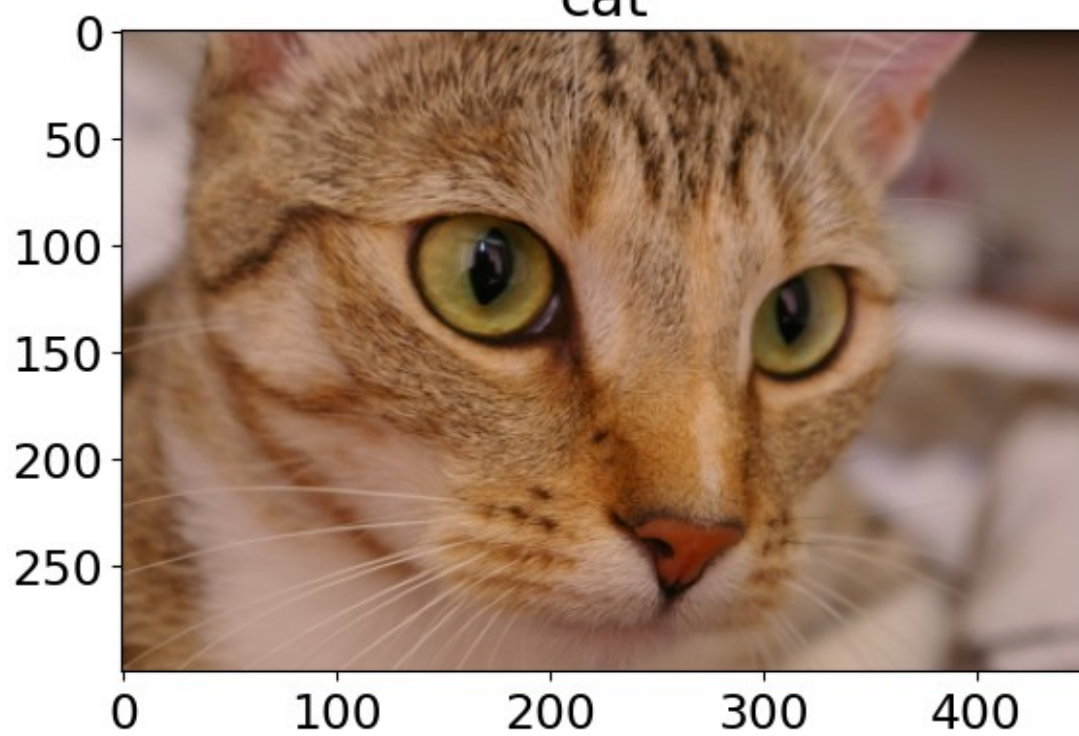


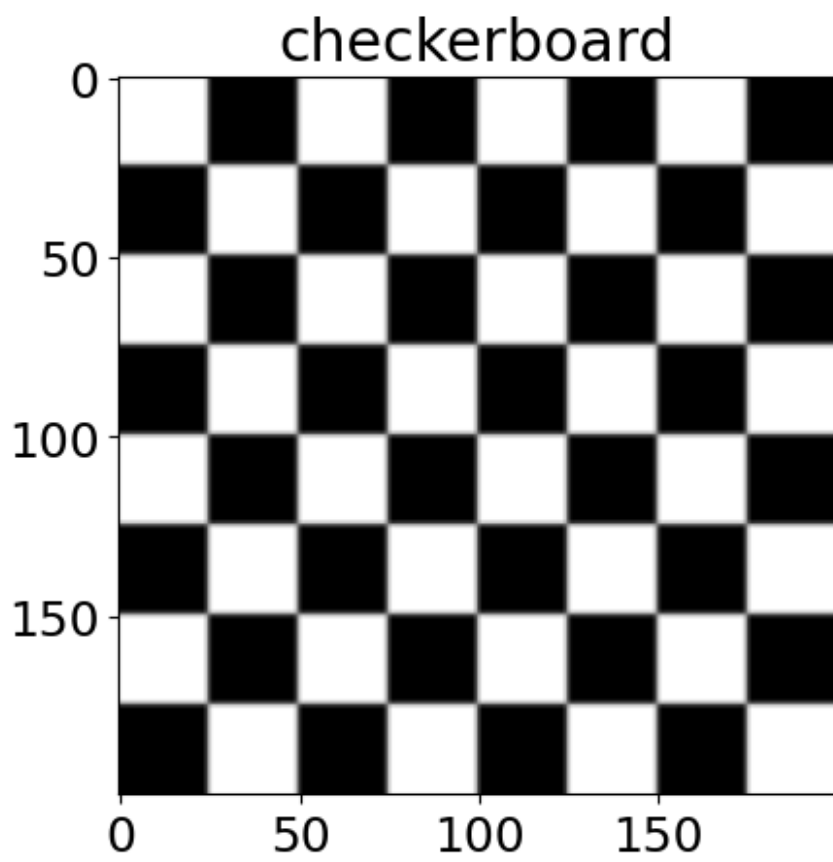


camera

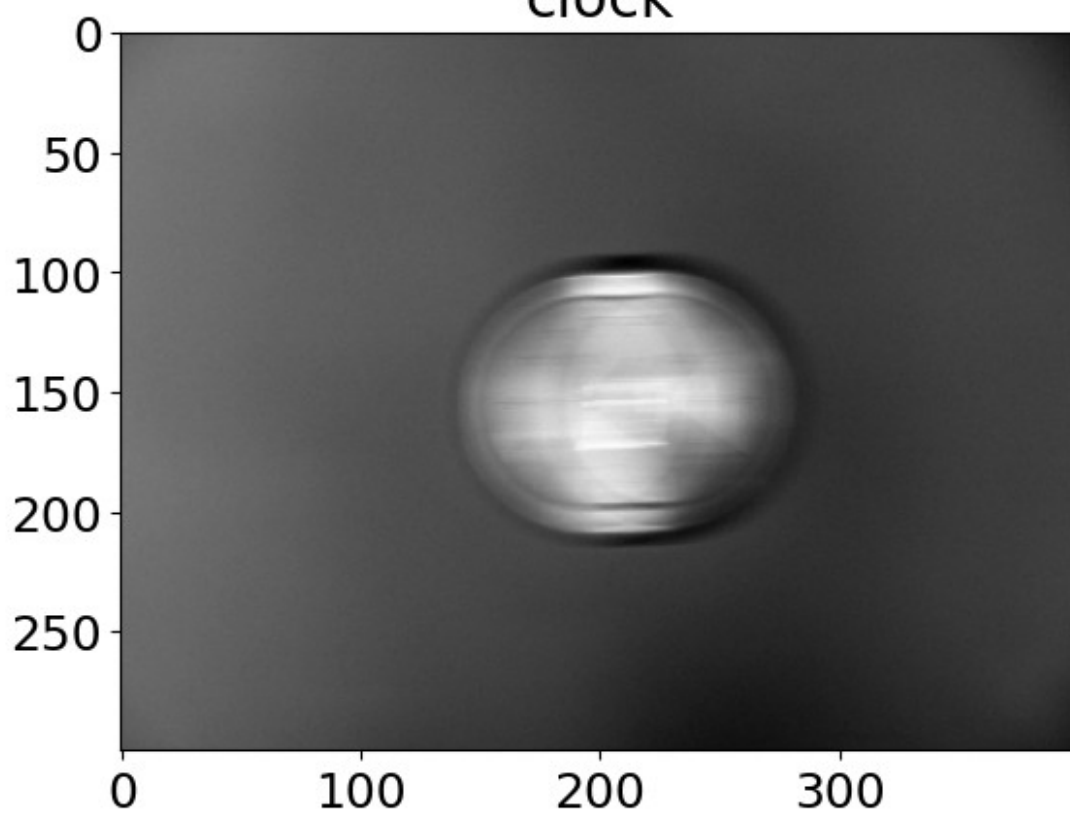


cat

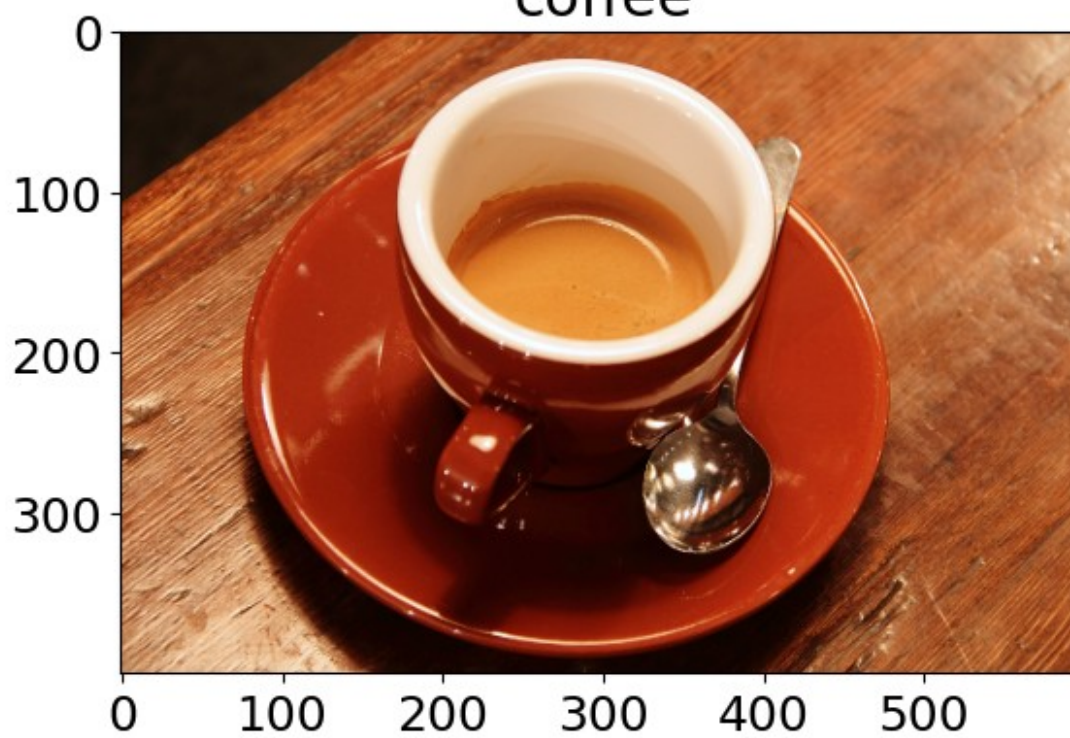




clock

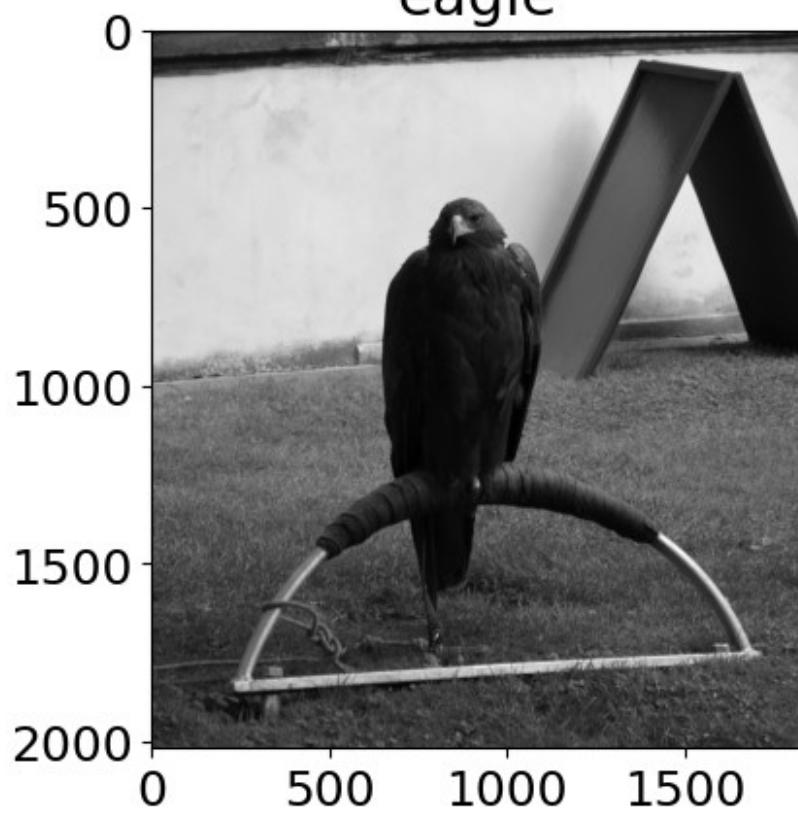


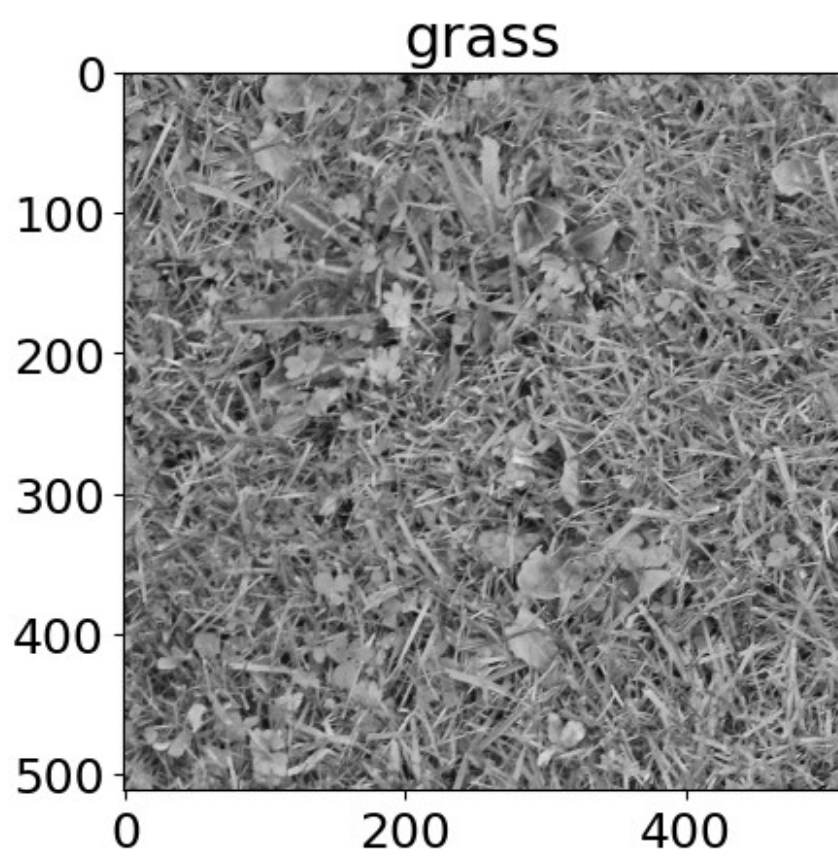
coffee

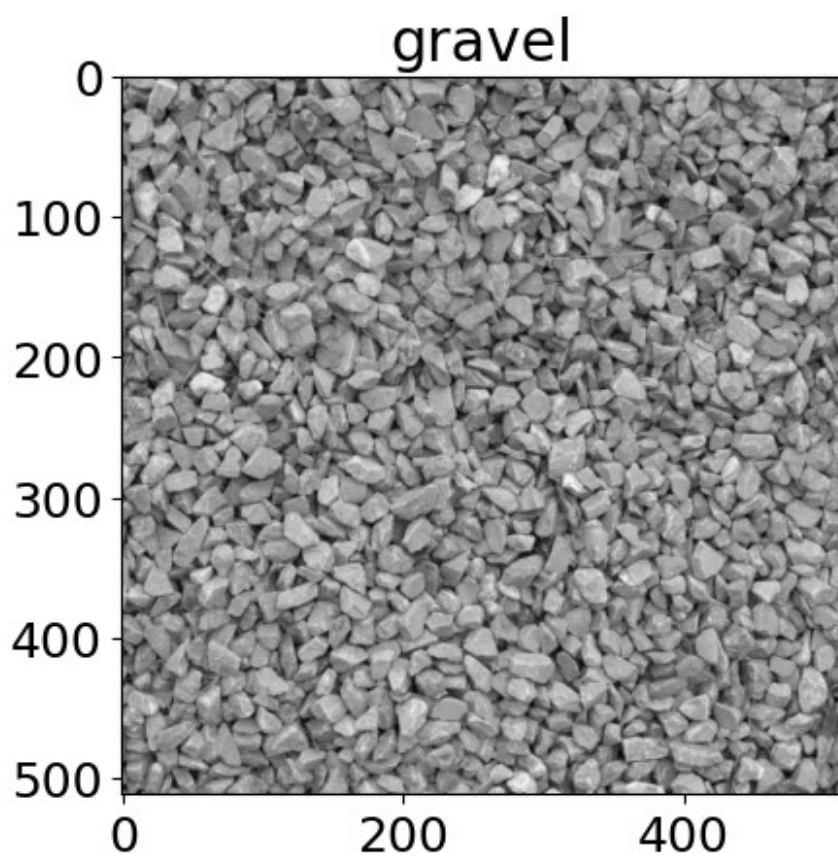


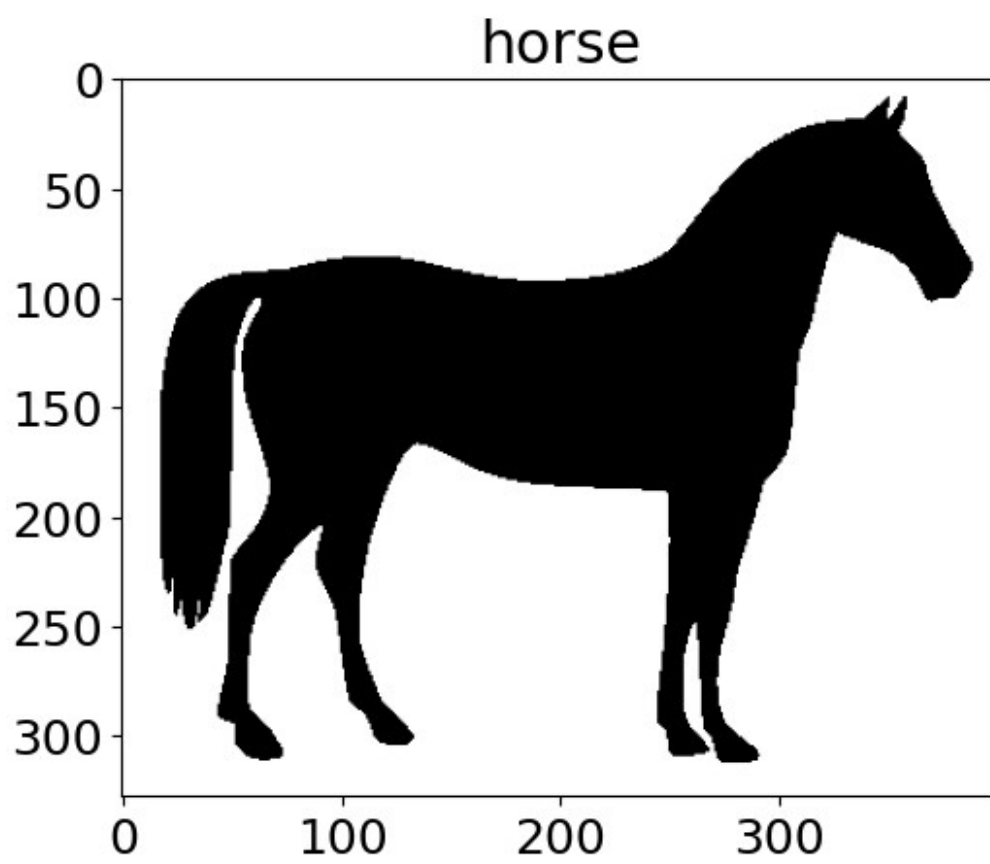
A 4x6 grid of 24 ancient coins, labeled "COINS" at the top. The grid is indexed by row (0 to 300) and column (0 to 300). The coins are arranged in a grid, with some showing obverse and others reverse sides.

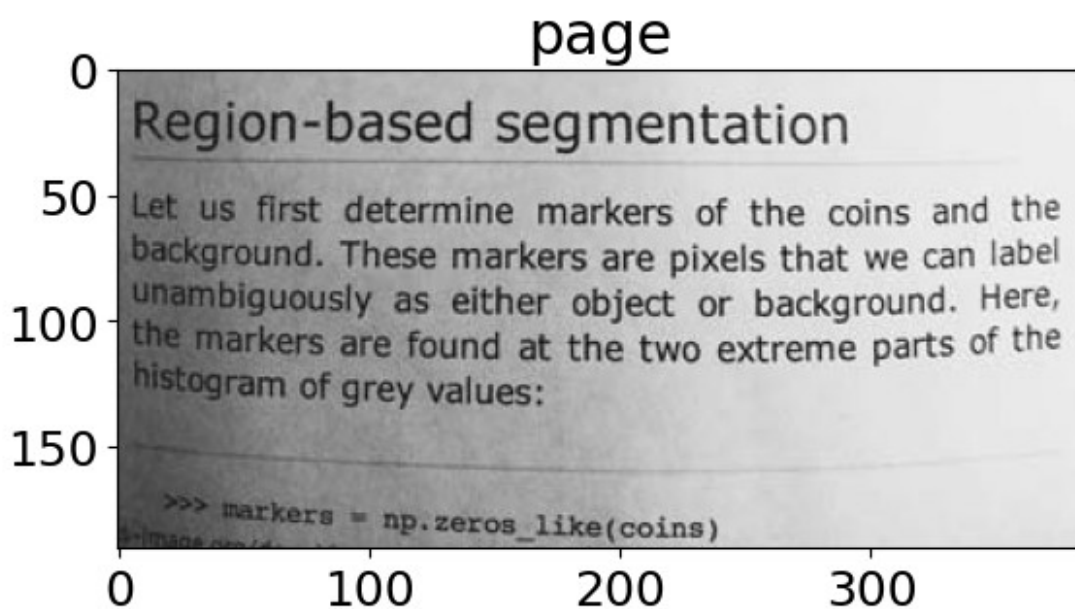
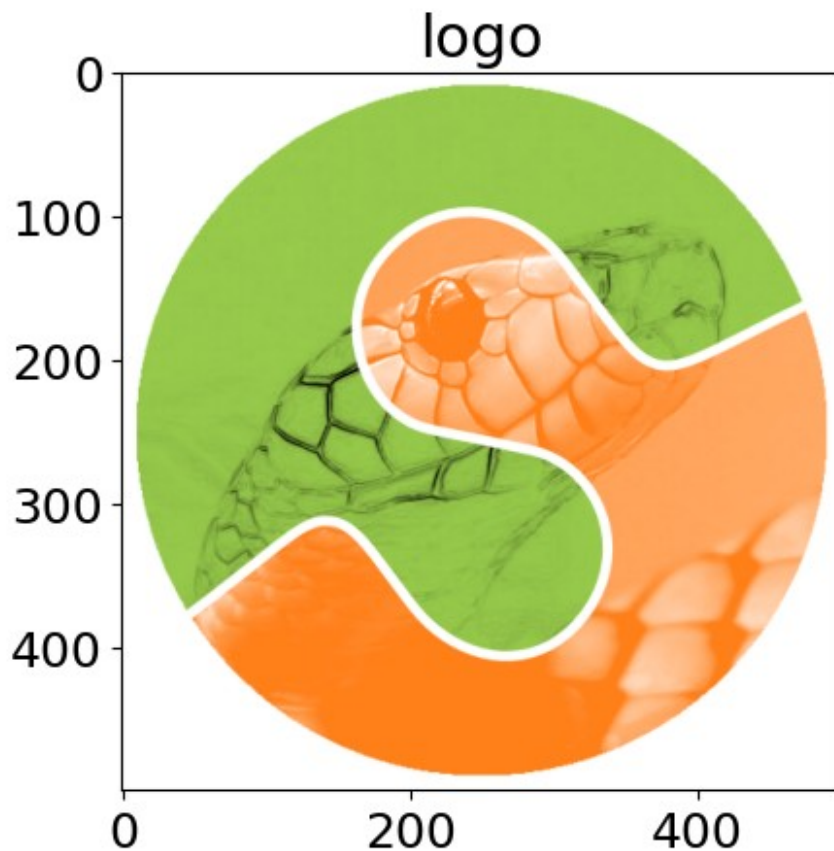
eagle

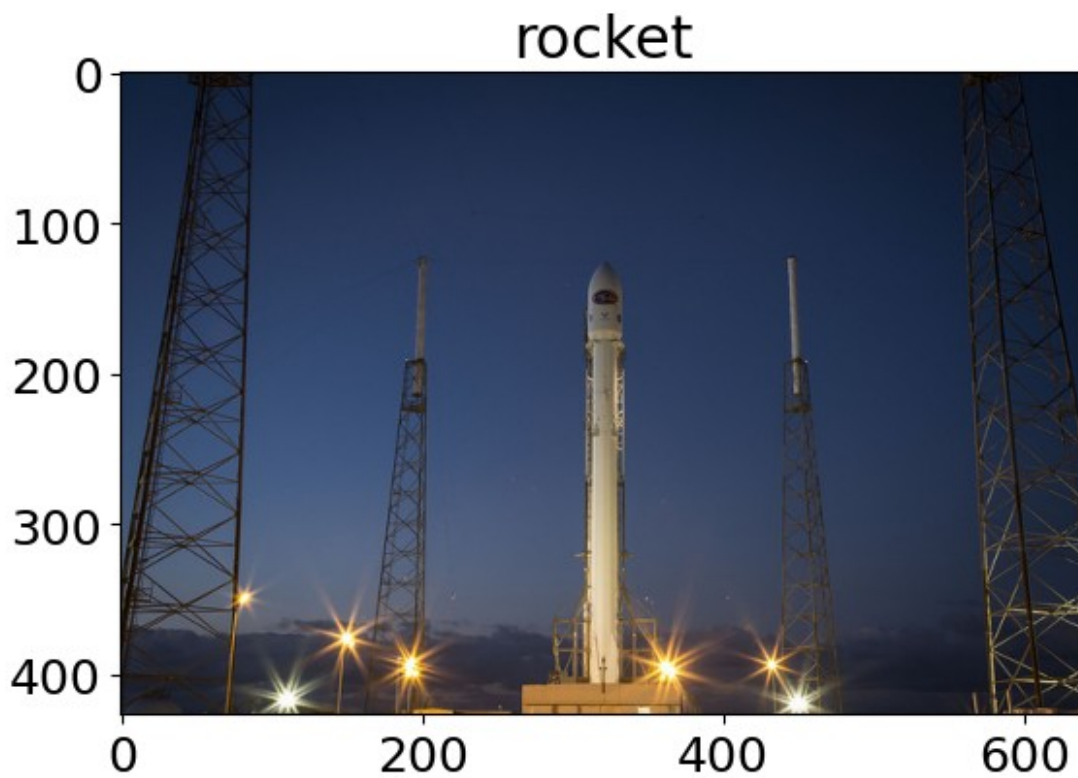
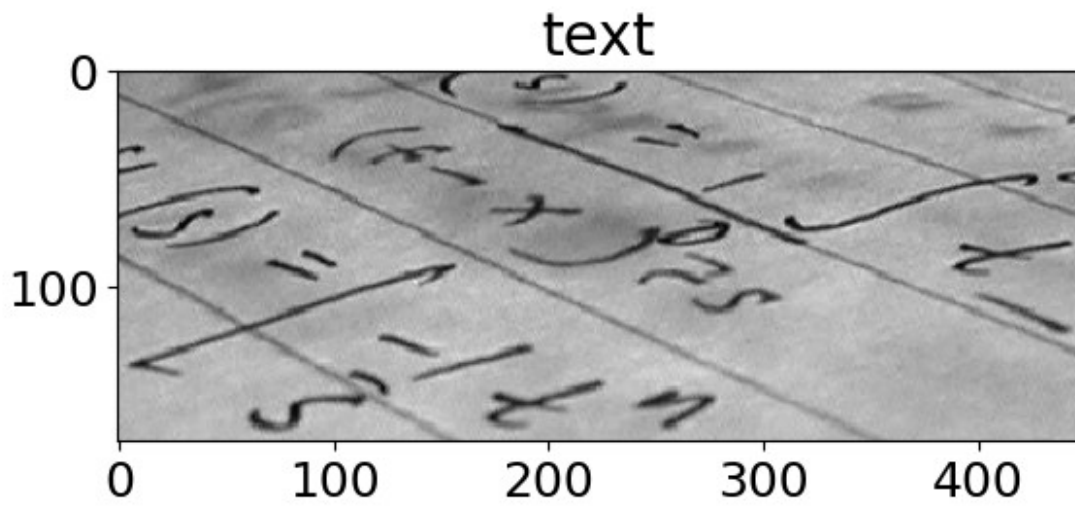












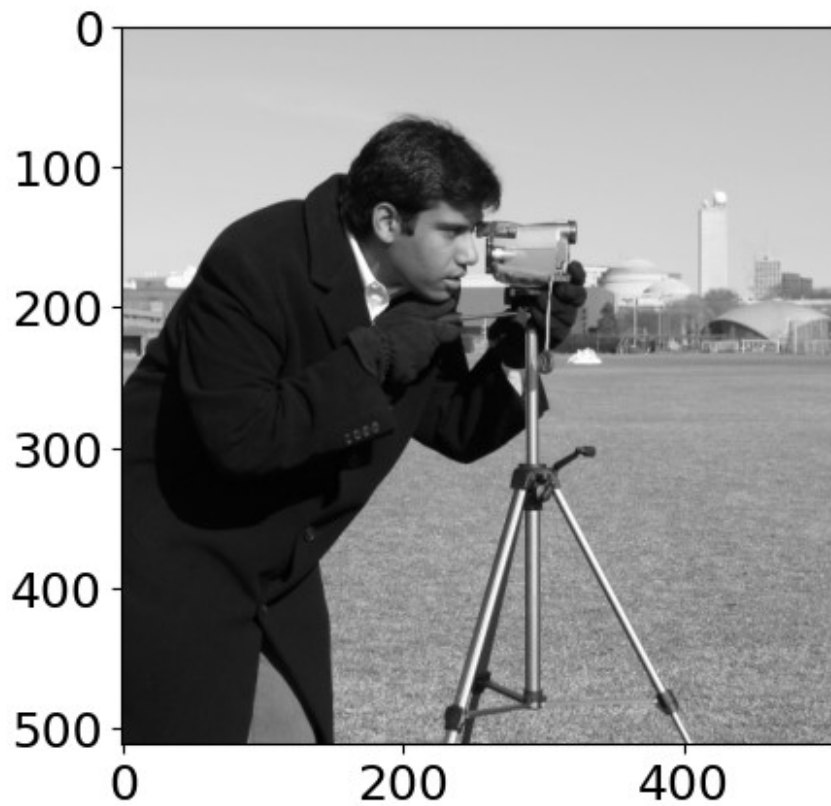
Task#2

a. Loading/reading an image using OpenCV

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread('images/camera.tif')
```

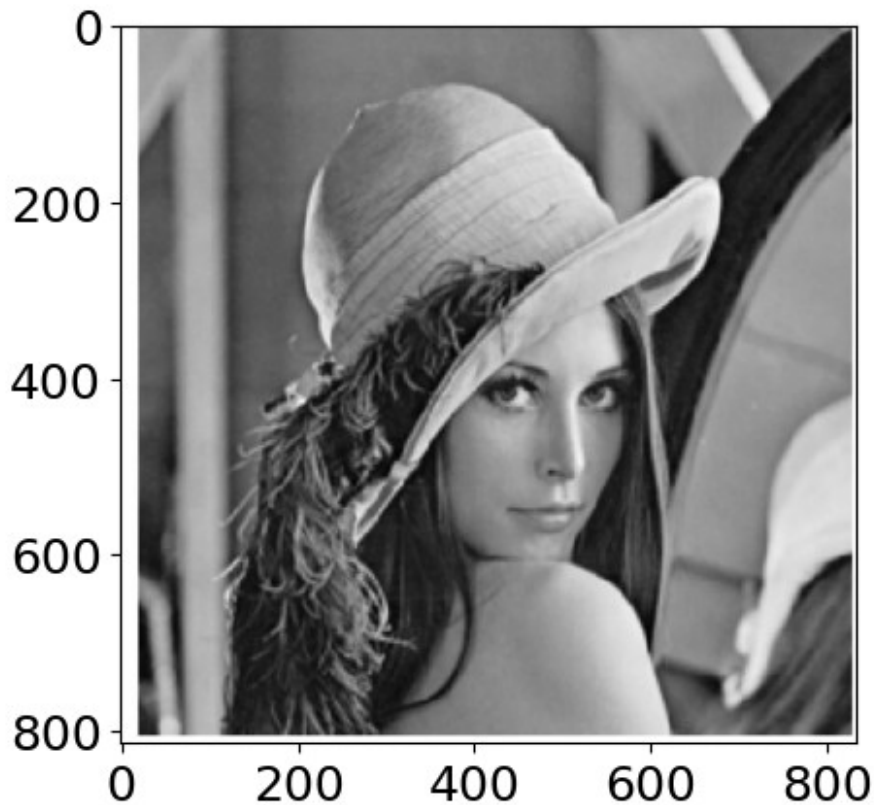
```
plt.imshow(img)
plt.show()
```



b. Loading/reading image using PIL

```
from PIL import Image
import matplotlib.cm as cm

img2 = Image.open('images/lena_gray_256.tif')
plt.imshow(img2, cmap = cm.Greys_r)
plt.show()
```



Task#3

a. Saving image to disk using OpenCV

```
cv2.imwrite('new_image.jpg', img)
```

True

b. Saving an image to disk using PIL

```
img2.save('new_image2.jpg')
```

Task#4

```
print(img.shape)
print(img)
import numpy as np
img_array = np.array(img2)
print(img_array.shape)
print(img_array)

(512, 512, 3)
[[[200 200 200]
  [200 200 200]
```

```

[200 200 200]
...
[189 189 189]
[190 190 190]
[190 190 190]]

[[200 200 200]
 [199 199 199]
 [199 199 199]
 ...
 [190 190 190]
 [190 190 190]
 [190 190 190]]

[[199 199 199]
 [199 199 199]
 [199 199 199]
 ...
 [190 190 190]
 [190 190 190]
 [190 190 190]]

...

[[ 25  25  25]
 [ 25  25  25]
 [ 27  27  27]
 ...
 [139 139 139]
 [122 122 122]
 [147 147 147]]

[[ 25  25  25]
 [ 25  25  25]
 [ 26  26  26]
 ...
 [158 158 158]
 [141 141 141]
 [168 168 168]]

[[ 25  25  25]
 [ 25  25  25]
 [ 27  27  27]
 ...
 [151 151 151]
 [152 152 152]
 [149 149 149]]]
(814, 834, 3)
[[[255 255 255]
 [255 255 255]

```

```
[255 255 255]
...
[255 255 255]
[255 255 255]
[255 255 255]]
```

```
[ [255 255 255]
  [255 255 255]
  [255 255 255]
  ...
  [255 255 255]
  [255 255 255]
  [255 255 255]]
```

```
[ [255 255 255]
  [255 255 255]
  [255 255 255]
  ...
  [247 247 247]
  [255 255 255]
  [255 255 255]]
```

...

```
[ [255 255 255]
  [255 255 255]
  [255 255 255]
  ...
  [255 255 255]
  [255 255 255]
  [255 255 255]]
```

```
[ [255 255 255]
  [255 255 255]
  [255 255 255]
  ...
  [255 255 255]
  [255 255 255]
  [255 255 255]]
```

```
[ [255 255 255]
  [255 255 255]
  [255 255 255]
  ...
  [255 255 255]
  [255 255 255]
  [255 255 255]]]
```